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# ABSTRACT

Accurate and real-time prediction of transient stability is essential for ensuring the secure operation of power systems under large disturbances. The Transient Energy Function (TEF) method offers a fast and analytical way to assess stability by comparing system energy to a critical threshold, but its effectiveness depends heavily on precise knowledge of generator rotor angles and speeds. This project proposes an LSTM-based approach to enhance the accuracy of generator state prediction during fault conditions. By training a Long Short-Term Memory (LSTM) neural network on time-series simulation data from the IEEE 9-bus system, we aim to predict high-fidelity generator angles and speeds in real time. These predicted states are then integrated into the TEF framework to calculate kinetic and potential energies, thereby enabling more accurate estimation of the transient stability margin ($\Delta V$). The proposed method is expected to outperform conventional approaches by providing more reliable stability assessment and improving control action decisions such as generation shedding.

## INTRODUCTION/BACKGROUND

Transient stability is a critical aspect of power system operation, concerning the ability of synchronous generators to maintain synchronism after large disturbances like short circuits or line tripping. Failure to maintain synchronism can lead to cascading outages and widespread blackouts. Real-time prediction and control of transient stability are vital for ensuring grid reliability and preventing such catastrophic events.

**Understanding the Key Terms:**

**Transient Stability Assessment:** This refers to the evaluation of a power system's ability to remain in synchronism after a large disturbance (e.g., a short circuit). It involves determining if generators will "fall out of step" with each other, which can lead to system collapse.

**High-Fidelity Generator State Prediction:** In the context of this project, "high-fidelity" means highly accurate and precise prediction of dynamic generator variables, specifically rotor angles ($\delta$) and speeds ($\omega$). These states are crucial for understanding the stability of the system.

**Transient Energy Function (TEF):** A direct method for transient stability analysis. Instead of simulating the entire post-fault trajectory (which can be time-consuming), TEF provides a scalar energy function that quantifies the system's "energy" during a disturbance. The concept relies on comparing the system's kinetic and potential energy to a critical energy value (related to the Controlling Unstable Equilibrium Point), thereby providing a direct measure of stability margin ($\Delta V$).

**LSTM-Based Approach:** This refers to the use of Long Short-Term Memory (LSTM) networks, a specialized type of Recurrent Neural Network (RNN). LSTM is a powerful machine learning technique particularly designed to handle and learn from sequential data (time-series data), making them ideal for predicting dynamic system states like generator angles and speeds over time. Unlike simpler neural networks, LSTM have "memory cells" that allow them to remember information for long periods, which is crucial for capturing the complex dynamics of a power system during transients[3]. In our project, the LSTM will be trained on historical generator state data from various fault scenarios. Once trained, it will learn the complex, non-linear relationships between past and future states. This allows it to predict the generator angles and speeds with high accuracy given a sequence of past observations. This is a *regression* task, where the output is a continuous numerical value (angle and speed), not a classification (stable/unstable).

**Phasor Measurement Unit (PMU):** Advanced devices that provide synchronized voltage and current phasor measurements from widely dispersed locations in the power grid. PMU offers high-resolution, time-stamped data, which is essential for real-time applications like transient stability assessment and wide-area monitoring. The base paper heavily relies on PMU data.

**Controlling Unstable Equilibrium Point (CUEP):** In TEF analysis, the CUEP is a specific unstable equilibrium point on the system's energy surface. It represents a "saddle point" or a "mountain pass" that the system's trajectory must cross if it is to become unstable. Accurately identifying the CUEP is critical because it defines the critical energy boundary for transient stability. If the system's transient energy exceeds the energy at the CUEP, it becomes unstable. Determining the CUEP in real-time is computationally challenging, and the base paper proposes techniques to overcome this.

**Modes of Disturbance (MOD):** In the base paper, MOD refers to specific patterns of system instability that generators exhibit after a fault. Different fault locations and severities can lead to different groups of generators losing synchronism together. The paper pre-classifies these patterns offline.

**Boundary Controlling Unstable Equilibrium Point (BCU) Method:** This is a technique used in the base paper to find the CUEP. It involves identifying the unstable equilibrium point on the stability boundary that is closest to the fault-on trajectory or the exit point from the stable region. The paper aims to simplify this real-time identification using normalized kinetic and potential energies.

The Transient Energy Function (TEF) method, as detailed in the base paper "Real Time Prediction and Control of Transient Stability Using Transient Energy Function" by P. Bhui and N. Senroy, offers a quantitative measure of stability margin using PMU data. However, a crucial aspect for accurate TEF application is the precise knowledge and prediction of generator rotor angles and speeds. The identified research gap is that the accuracy of generator angle and speed prediction in existing TEF applications can be improved, which directly impacts the reliability of stability assessment and control.

This project proposes to address this limitation by integrating Long Short-Term Memory (LSTM) neural networks to provide high-fidelity predictions of generator states. This integration aims to enhance the overall accuracy and reliability of transient stability assessment and control within the TEF framework, specifically focusing on the well-known IEEE 9-bus system as a case study.

# OBJECTIVES

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**Primary Objective:**

- To develop and implement an LSTM (Long Short-Term Memory) neural network model for accurate, real-time prediction of generator angles and speeds in the IEEE 9-bus system following a fault

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**Secondary Objectives:**

- To integrate the high-fidelity generator state predictions into the Transient Energy Function (TEF) framework for improved transient stability assessment
- To evaluate the effectiveness of the LSTM-based prediction in enhancing the accuracy of the Transient Stability Margin (TSM) calculation and control action determination (e.g., generation shedding) within the TEF methodology
- To demonstrate the applicability of the enhanced TEF method on the IEEE 9-bus test system

## LITERATURE REVIEW

The current literature extensively covers transient stability assessment methods. Traditional approaches include time-domain simulations and direct methods like the Transient Energy Function (TEF). TEF, as highlighted in the base paper [1], provides a fast and quantitative measure of stability. However, its real-time application has been challenged by difficulties in accurately determining the Controlling Unstable Equilibrium Point (CUEP) and the sensitivity to accurate generator state (angles and speeds) measurements or predictions. The base paper specifically addresses the CUEP challenge through a combination of offline analysis (MOD) and online matching techniques (BCU method).

The specific challenge of accurate time-series prediction, such as generator angles and speeds, has led to the adoption of recurrent neural network (RNN). LSTM, a specialized type of RNN, are particularly effective at capturing long-term dependencies in sequential data, making them suitable for dynamic system prediction. Several studies have applied LSTM in power systems for load forecasting [2], renewable energy generation prediction, and fault detection. However, their specific application for high-fidelity real-time generator state prediction to directly enhance TEF-based transient stability assessment, especially in addressing the accuracy of angles and speeds, represents a significant research opportunity and the core focus of this proposal.

## METHODOLOGY

Our methodology will build upon the foundation of the base paper [1] and integrate an LSTM model for improved generator state prediction. In the original paper combination of quadratic and cubic curve fitting was used to predict the rotor angle and speed. We will be using LSTM machine learning algorithm to predict the rotor angle and speed corresponding to critical MOD obtained. We will specifically focus on the IEEE 9-bus test system.

### A. Overview of the Original Paper's Methodology

The original paper proposes a two-phase approach for real-time transient stability prediction and control: an **Offline Phase** and an **Online Phase**.

**Offline Phase (Original Paper):**

1. **Fault Simulation and Data Collection:** Numerous fault scenarios (locations, types, durations) are simulated offline on the power system model. For each scenario, time-domain simulation data, including generator angles and speeds, power flows, and bus voltages, are collected.

2. **Identification of Modes of Disturbance (MOD):** For each fault, the system's post-fault behavior is analyzed to identify specific "Modes of Disturbance." This involves identifying which groups of generators lose synchronism (swing together) or accelerate/decelerate in particular patterns. This creates a catalog or look-up table of MODs for different fault conditions and post-fault topologies.

3. **CUEP Calculation and Critical Energy:** For each identified MOD, the corresponding CUEP is determined offline using iterative techniques or continuation methods. The critical transient energy corresponding to this CUEP is also calculated and stored in the look-up table. This step is computationally intensive and done offline.

4. **Normalized Kinetic and Potential Energy Reference:** The paper calculates normalized kinetic energy (Tnorm) and normalized potential energy (ΔVPE,norm) for various fault scenarios and stores them in the look-up table. These normalized values help in online matching.

**Online Phase (Original Paper):**

1. **PMU Data Acquisition:** After an actual fault occurs in the real power system, real-time generator angles and speeds (and potentially other data) are acquired from PMUs.
2. **Kinetic Energy Calculation:** The online PMU data is used to calculate the kinetic energy of the system during the transient.

3. **MOD Ranking and Selection:** The calculated online kinetic energy is "normalized" and compared with the offline pre-computed normalized kinetic energies in the look-up table. This step helps in ranking the most probable MODs that match the observed real-time behavior.
4. **CUEP Refinement (using Normalized Potential Energy Margin):** From the ranked MODs, the correct MOD is selected based on the lowest normalized potential energy margin. This step is crucial for accurate CUEP identification in real-time, avoiding the need for complex iterative calculations. The paper aims to overcome the real-time CUEP finding difficulty through this lookup and matching process.
5. **Transient Stability Margin (TSM) Calculation:** Once the CUEP and critical energy are determined for the identified MOD, the Transient Stability Margin ($\Delta V$) is calculated by subtracting the critical energy from the transient energy at fault clearing.
6. **Control Action Determination:** Based on the calculated TSM, appropriate control actions (e.g., generation shedding for unstable cases, or preventive measures for critically stable cases) are determined to stabilize the system.

## B. Our Proposed Methodology (with LSTM Integration)

Our methodology will build upon the foundation of the base paper and primarily modify the "Data Generation and Pre-processing" and "Real-time Prediction" aspects to integrate an LSTM model for improved generator state prediction. We will specifically focus on the IEEE 9-bus test system.

### 1. Data Generation and Pre-processing

a. **IEEE 9-Bus System Modeling:** Model the IEEE 9-bus system in a suitable power system simulation environment (e.g., MATLAB/Simulink with Simscape Power Systems, or Python with libraries like PyPSA-Eur, pandapower). We will represent generators with classical models initially, then consider more detailed models if time permits, as the complexity of generator models affects the dynamic response.

b. **Fault Simulation:** Simulate various fault scenarios (e.g., three-phase faults, single-line-to-ground faults) at different locations (buses, lines) and with varying fault durations. This data generation will be more extensive than just for MOD identification, as the LSTM requires rich, diverse time-series data for robust training.

c. **Data Collection:** Collect high-resolution time-series data for generator rotor angles ($\delta$) and speeds ($\omega$) for all generators in the IEEE 9-bus system *during and after* each fault. This data will serve as the training and testing dataset specifically for the LSTM model.

   d. **Data Normalization:** Normalize the collected data to a consistent range (e.g., 0 to 1 or -1 to 1) to improve LSTM training performance and prevent numerical issues.

2. **LSTM Model Development**
   a. **LSTM Architecture Design:** Design an appropriate LSTM network architecture. This will involve determining the number of LSTM layers (e.g., 2-3 layers), the number of neurons per layer (e.g., 50-100 units), the input sequence length (how many past time steps are fed into the LSTM, e.g., 10-20 previous data points), and the output prediction horizon (how many future time steps are predicted, e.g., 1-5 future points). We will use Python with libraries like TensorFlow or Keras.
   b. **Training Data Preparation:** Divide the collected time-series data into sequences suitable for LSTM training. Each input sequence will consist of historical generator angle and speed data, and the corresponding output will be the future angle and speed values. For instance, if we want to predict the next 5 time steps, an input sequence of 10 previous time steps would be mapped to a target output of 5 future time steps.
   c. **Model Training:** Train the LSTM model using the prepared dataset. We will use a suitable optimization algorithm (e.g., Adam, RMSprop) and a loss function appropriate for regression tasks (e.g., Mean Squared Error (MSE) or Mean Absolute Error (MAE)). The training process involves feeding the input sequences to the LSTM and adjusting its internal weights to minimize the prediction error.
   d. **Hyperparameter Tuning:** Experiment with different hyperparameters (learning rate, batch size, number of epochs, network architecture, dropout rates) to optimize the LSTM model's prediction accuracy and prevent overfitting. This iterative process is crucial for achieving high-fidelity predictions.

3. **Integration with Transient Energy Function (TEF)**
   a. **Real-time Prediction Simulation (Enhanced Online Phase):** Instead of relying solely on immediate PMU data at the fault clearing instant to select MOD and CUEP, we will use the trained LSTM model. During simulated fault conditions, the LSTM will take the recent PMU-like data (generator angles and speeds) as input and predict the *future* trajectory of these states, particularly at and immediately after the fault clearing instant. This provides a more accurate and dynamic understanding of the system's impending behavior.
   b. **TEF Calculation (with LSTM-predicted States):** Utilize these predicted high-fidelity generator states (angles and speeds) within the TEF framework. The improved accuracy of these predicted states will directly feed into the calculation of the system's kinetic and potential energies,

leading to a more precise calculation of the transient energy margin ($\Delta V$) as described in the base paper (Equation 2).

c. **CUEP Calculation:** While the base paper focuses on MOD and BCU techniques for CUEP determination based on historical data, our LSTM-predicted states can potentially provide a more accurate "snapshot" or short-term trajectory of the system's state close to the stability boundary. This could assist or refine the CUEP identification process, potentially leading to more accurate CUEP identification without extensive offline simulations for every specific scenario, or validating the CUEP found by the MOD/BCU methods.
d. **Control Action Determination:** Based on the more accurately calculated transient stability margin (derived from LSTM-enhanced state prediction), determine necessary control actions, such as generation shedding, following the principles outlined in the base paper. The improved accuracy of the stability margin allows for more optimal and timely control interventions.

4. **Performance Evaluation**
   a. **Transient Stability Assessment Accuracy:** Compare the transient stability assessment results (stable/unstable classification, exact stability margin values) obtained using the LSTM-predicted states with the actual ground truth from detailed time-domain simulations. This will validate the overall improvement in stability assessment.
   b. **Computational Efficiency:** Analyze the computational time required for LSTM prediction (inference time) and its integration into the TEF framework compared to traditional time-domain simulations for similar accuracy levels. We aim for a balance between accuracy and speed for real-time applicability.

## EXPECTED RESULTS

We expect to achieve the following results from this project:

**Improved Prediction Accuracy:**

**LSTM-Based Approach:** We expect to achieve significantly higher accuracy in predicting future generator angles and speeds. The LSTM's ability to learn complex temporal dependencies will allow it to capture the dynamic oscillatory behavior of generators more precisely than simple extrapolation or lookup methods. This "high-fidelity" prediction will directly translate to more accurate transient energy calculations.
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**Enhanced Transient Stability Assessment:**

**LSTM-Based Approach:** By providing more accurate and robust predictions of generator states, our method will lead to a more reliable and precise assessment of the transient stability margin ($\Delta V$). This improved accuracy will reduce the risk of misclassifying system stability (false positives or false negatives), leading to more confident and safer operation.

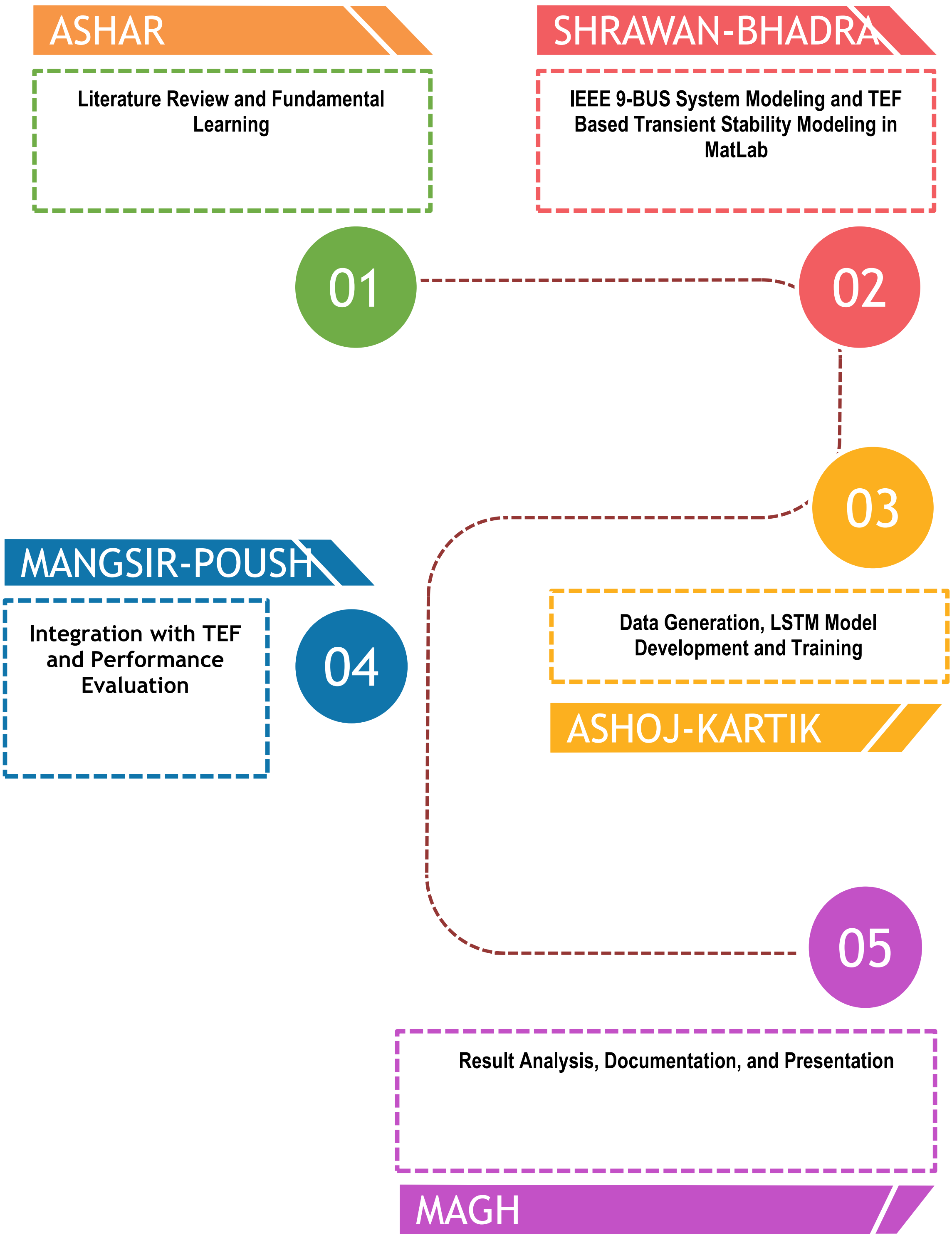
**Effective Control Action Determination (Our Method vs. Original Paper):**

**Original Paper's Approach:** The control actions are determined based on the TSM derived from the paper's methodology. If the TSM calculation has errors due to state estimation inaccuracies, the control actions might be suboptimal (e.g., shedding too much generation unnecessarily or failing to shed enough when needed).

**Our LSTM-Based Approach:** With a more accurate TSM calculation, we can determine more optimal and timely control actions. This means potentially smaller amounts of generation shedding for stable systems (economic benefit) or more effective shedding for unstable systems, contributing to faster and more reliable post-fault system recovery and preventing wider blackouts.

**Demonstration on IEEE 9-Bus System:** A fully implemented and validated system demonstrating the efficacy of the LSTM-based approach for transient stability assessment and control on the IEEE 9-bus test system. This will serve as a proof-of-concept for applying advanced ML techniques in power system dynamics.

## SCHEDULE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